



(12) **United States Patent**
Ripley

(10) **Patent No.:** **US 12,413,188 B2**
(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **SYSTEMS AND METHODS RELATED TO
POWER AMPLIFICATION AND POWER
SUPPLY CONTROL**

(71) Applicant: **SKYWORKS SOLUTIONS, INC.**,
Irvine, CA (US)

(72) Inventor: **David Steven Ripley**, Marion, IA (US)

(73) Assignee: **Skyworks Solutions, Inc.**, Irvine, CA
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/412,985**

(22) Filed: **Jan. 15, 2024**

(65) **Prior Publication Data**

US 2024/0223137 A1 Jul. 4, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/335,948, filed on
Jun. 1, 2021, now Pat. No. 11,876,492, which is a
(Continued)

(51) **Int. Cl.**
H04B 1/00 (2006.01)
H03F 1/02 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H03F 3/211** (2013.01); **H03F 1/0227**
(2013.01); **H03F 1/0261** (2013.01); **H03F**
1/565 (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC H03F 3/211; H03F 1/0227; H03F 1/0261;
H03F 1/565; H03F 3/191; H03F 3/195;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,913,970 B2 12/2014 Akhi et al.
2002/0123315 A1 9/2002 Hayashihara
(Continued)

FOREIGN PATENT DOCUMENTS

CN 101507105 8/2009
CN 201319559 9/2009
(Continued)

OTHER PUBLICATIONS

Application No. 201510639715.6, Power Amplifier Module With
Power Supply Control, filed Sep. 30, 2015.

(Continued)

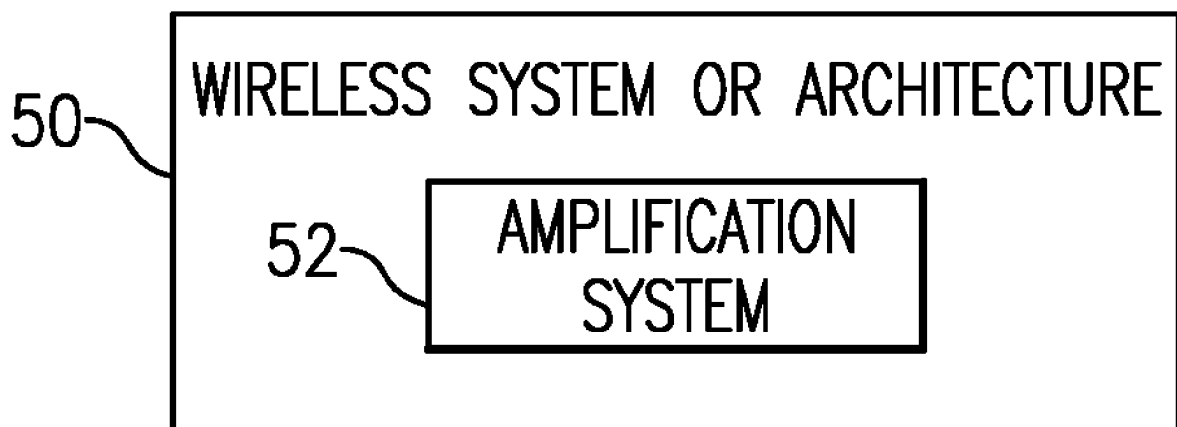
Primary Examiner — Ping Y Hsieh

(74) *Attorney, Agent, or Firm* — Chang & Hale LLP

(57) **ABSTRACT**

Systems and methods related to power amplification and
power supply control. A power amplification control system
can include an interface configured to receive a transceiver
control signal from a transceiver. The power amplification
control system can include a power amplifier control com-
ponent configured to generate a power amplifier control
signal based on the transceiver control signal from the
transceiver and a power supply control component config-
ured to generate a power supply control signal based on the
transceiver control signal from the transceiver and to gen-
erate the power supply control signal based on a local
control signal from the power amplifier control component.

20 Claims, 7 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/686,177, filed on Nov. 17, 2019, now Pat. No. 11,025,210, which is a continuation of application No. 14/867,209, filed on Sep. 28, 2015, now Pat. No. 10,483,926.

- (60) Provisional application No. 62/116,463, filed on Feb. 15, 2015.

(51) **Int. Cl.**

H03F 1/56 (2006.01)
H03F 3/191 (2006.01)
H03F 3/195 (2006.01)
H03F 3/21 (2006.01)
H04B 1/04 (2006.01)

(52) **U.S. Cl.**

CPC *H03F 3/191* (2013.01); *H03F 3/195* (2013.01); *H04B 1/0475* (2013.01); *H03F 2200/451* (2013.01); *H03F 2203/21142* (2013.01); *H04B 2001/0416* (2013.01)

(58) **Field of Classification Search**

CPC H03F 2200/451; H03F 2203/21142; H04B 1/0475; H04B 2001/0416
 See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2006/0178121 A1 8/2006 Hamalainen et al.
 2011/0298545 A1 12/2011 Morimoto et al.
 2012/0021704 A1* 1/2012 Chan H03F 1/02
 455/127.2
 2012/0214423 A1* 8/2012 Wallace H03F 1/0227
 455/69

2013/0207731 A1 8/2013 Balteanu
 2014/0002188 A1 1/2014 Chen et al.
 2014/0057684 A1* 2/2014 Khlal H03F 3/505
 455/574
 2014/0120854 A1* 5/2014 Briffa H03F 3/21
 455/127.3
 2014/0167856 A1 6/2014 Takahashi et al.
 2014/0266462 A1 9/2014 Schirmann et al.
 2014/0312974 A1 10/2014 Khesbak et al.

FOREIGN PATENT DOCUMENTS

CN 102104366 6/2011
 CN 104124929 10/2014
 JP 10336046 12/1998
 JP 2007019585 1/2007
 JP 2009100197 5/2009
 JP 2011130257 6/2011
 JP 2011259083 12/2011
 KR 100840399 6/2008
 KR 1020140121868 10/2014
 KR 1020150011851 2/2015
 WO 2013/039030 3/2013
 WO 2014/197399 12/2014

OTHER PUBLICATIONS

Application No. 201911120382.0, Power Amplifier Module With Power Supply Control, filed Nov. 15, 2019.
 Application No. 102015218920.1, Power Amplifier Module With Power Supply Control, filed Sep. 30, 2015.
 Application No. 2015-192573, Power Amplifier Module With Power Supply Control, filed Sep. 30, 2015.
 Application No. 10-2015-0138238, Power Amplifier Module With Power Supply Control, filed Sep. 30, 2015.
 Application No. 104132329, Power Amplifier Module With Power Supply Control, filed Sep. 30, 2015.

* cited by examiner

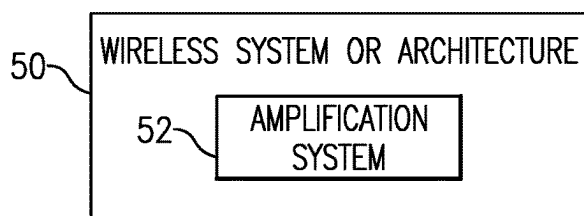


FIG. 1

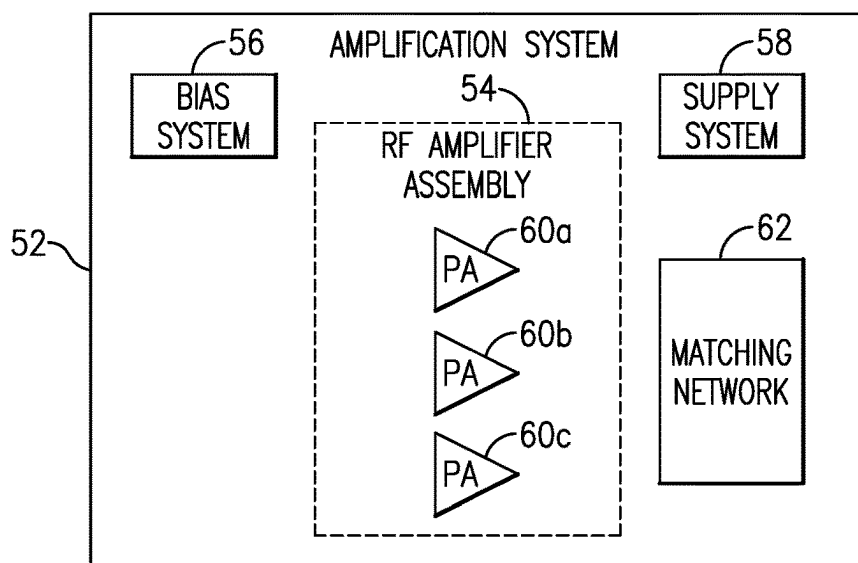


FIG. 2

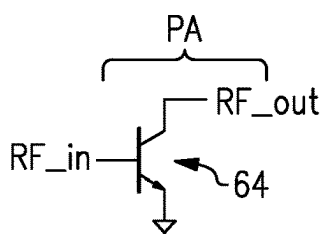


FIG. 3A

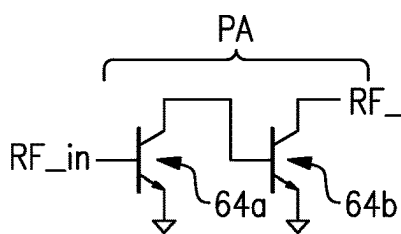


FIG. 3B

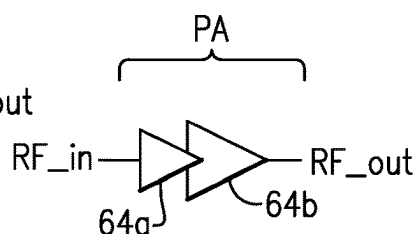


FIG. 3C

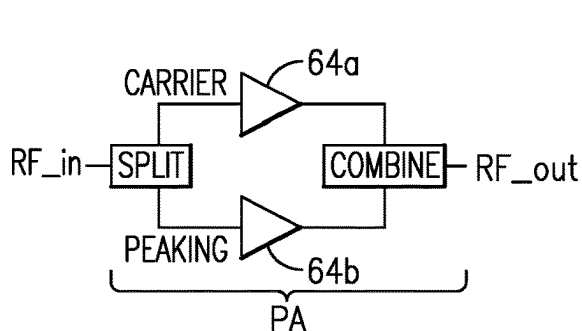


FIG. 3D

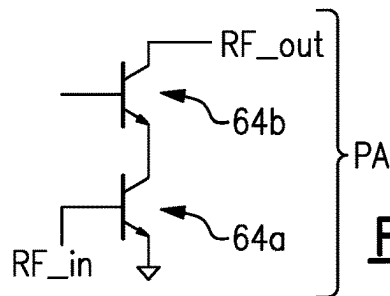
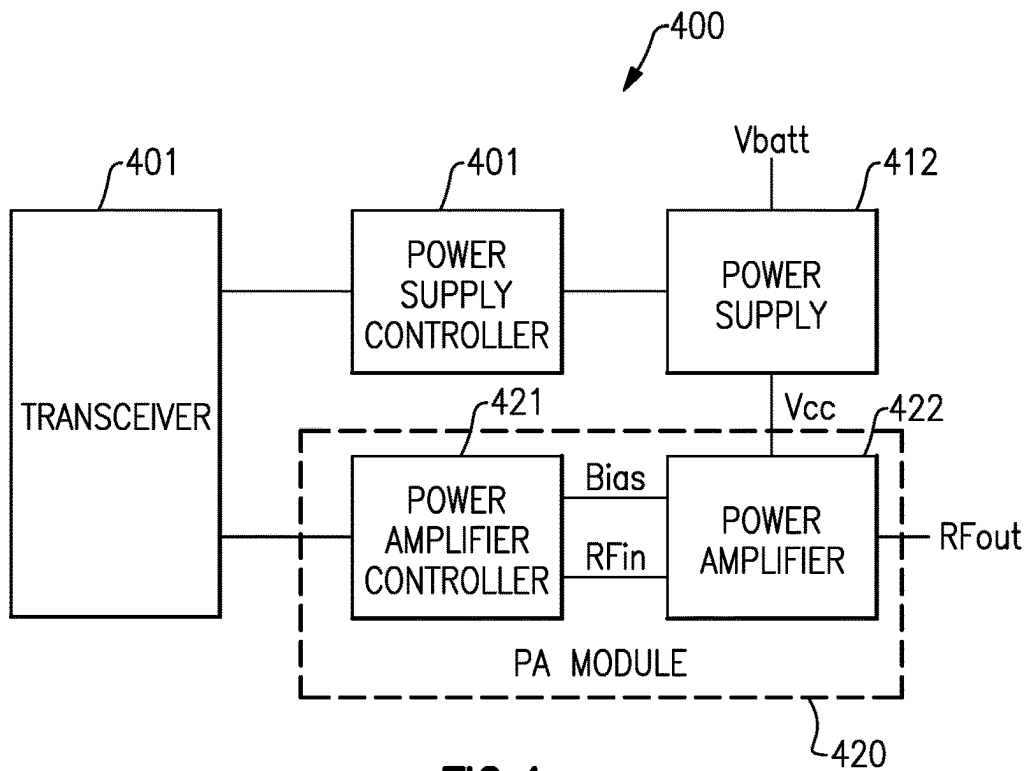
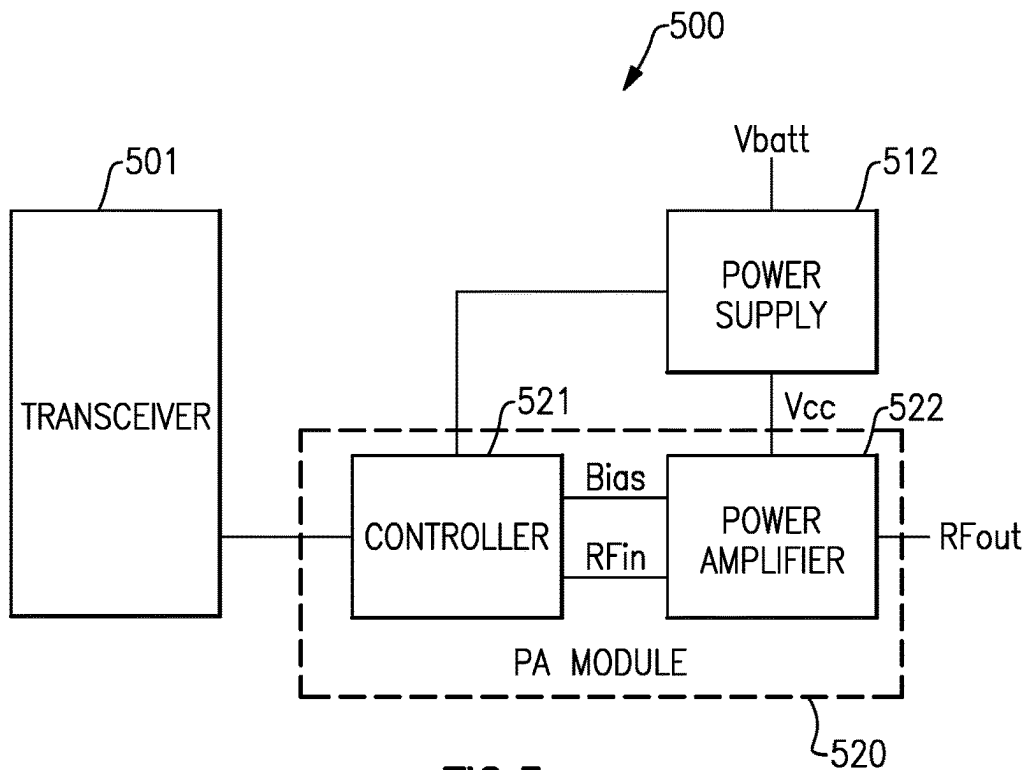
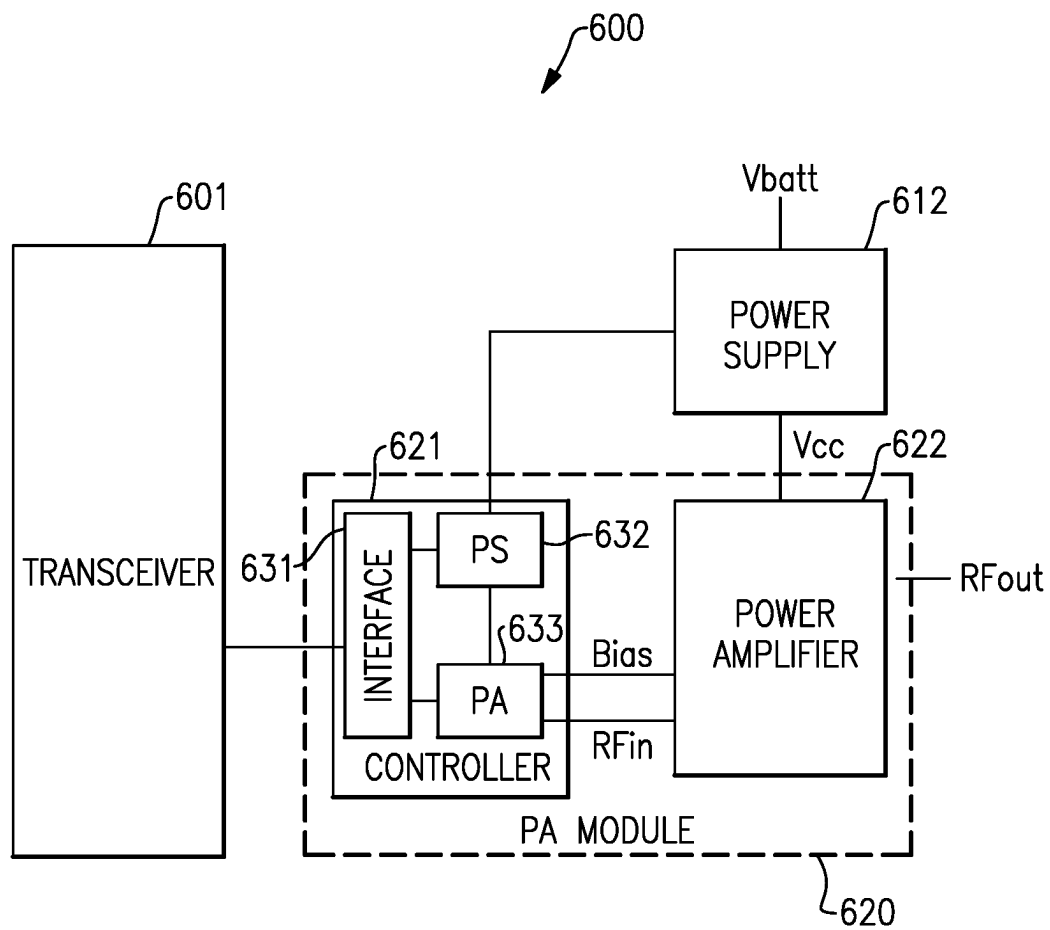


FIG. 3E

**FIG.4****FIG.5**

**FIG.6**

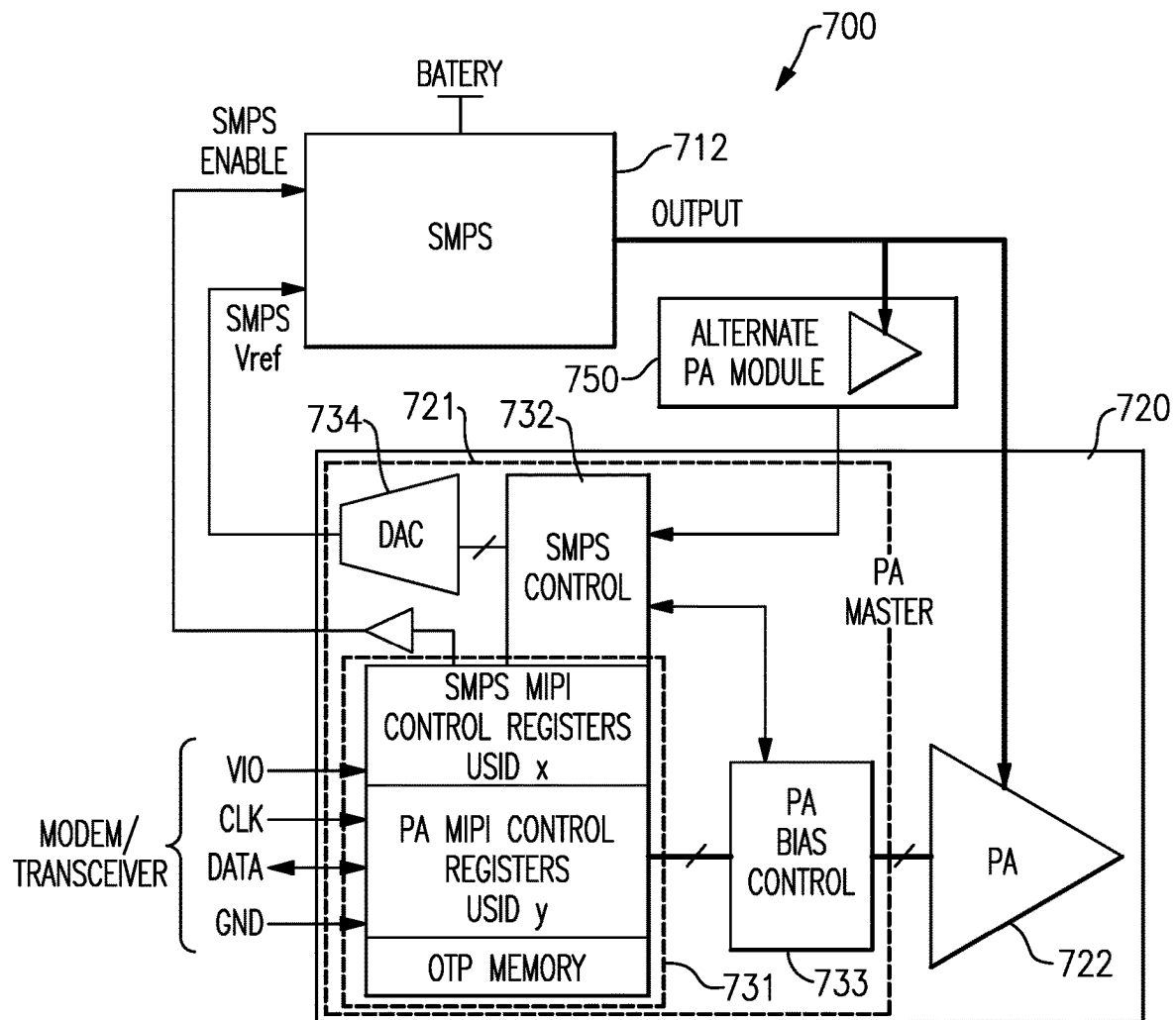


FIG.7

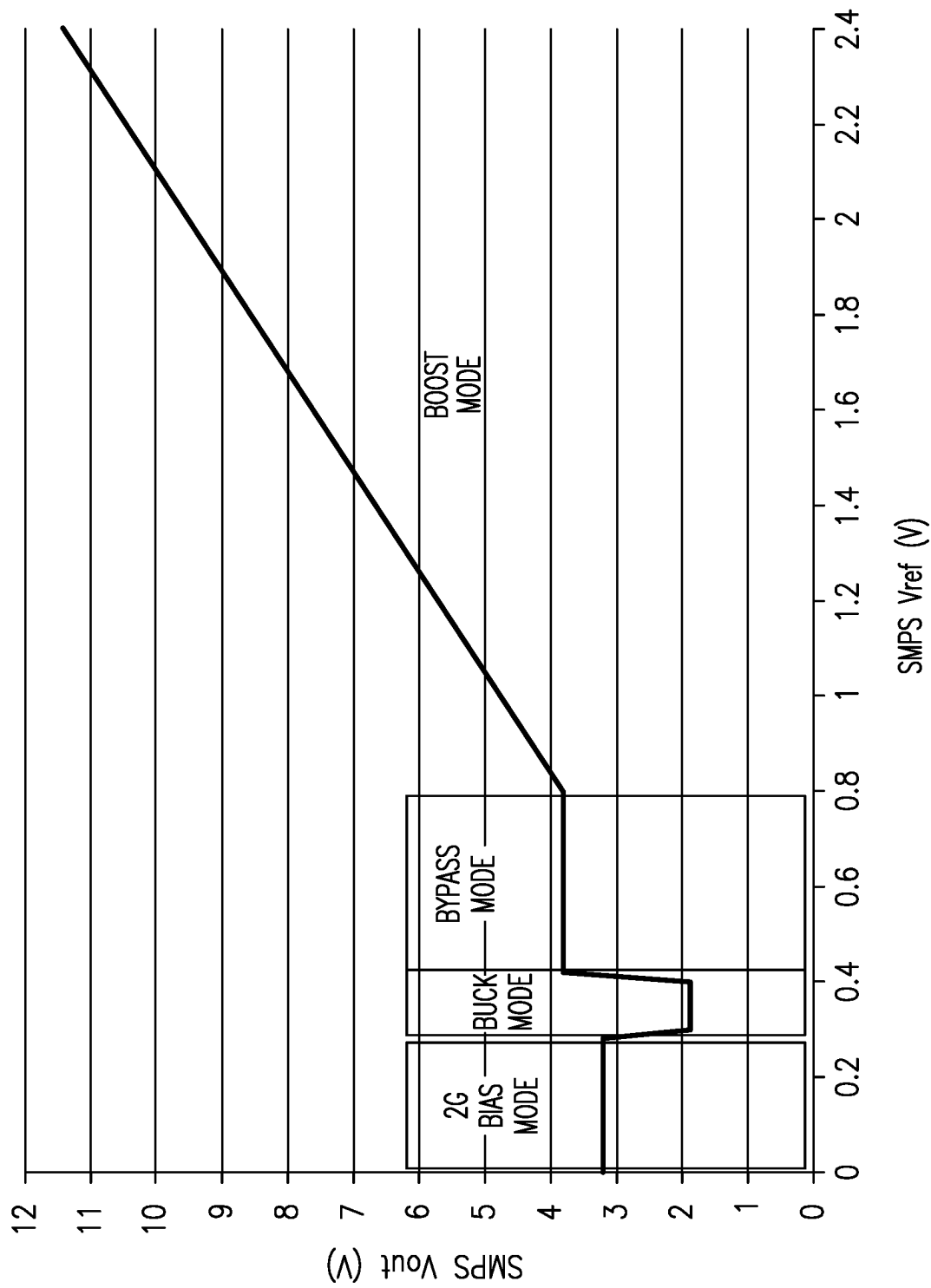
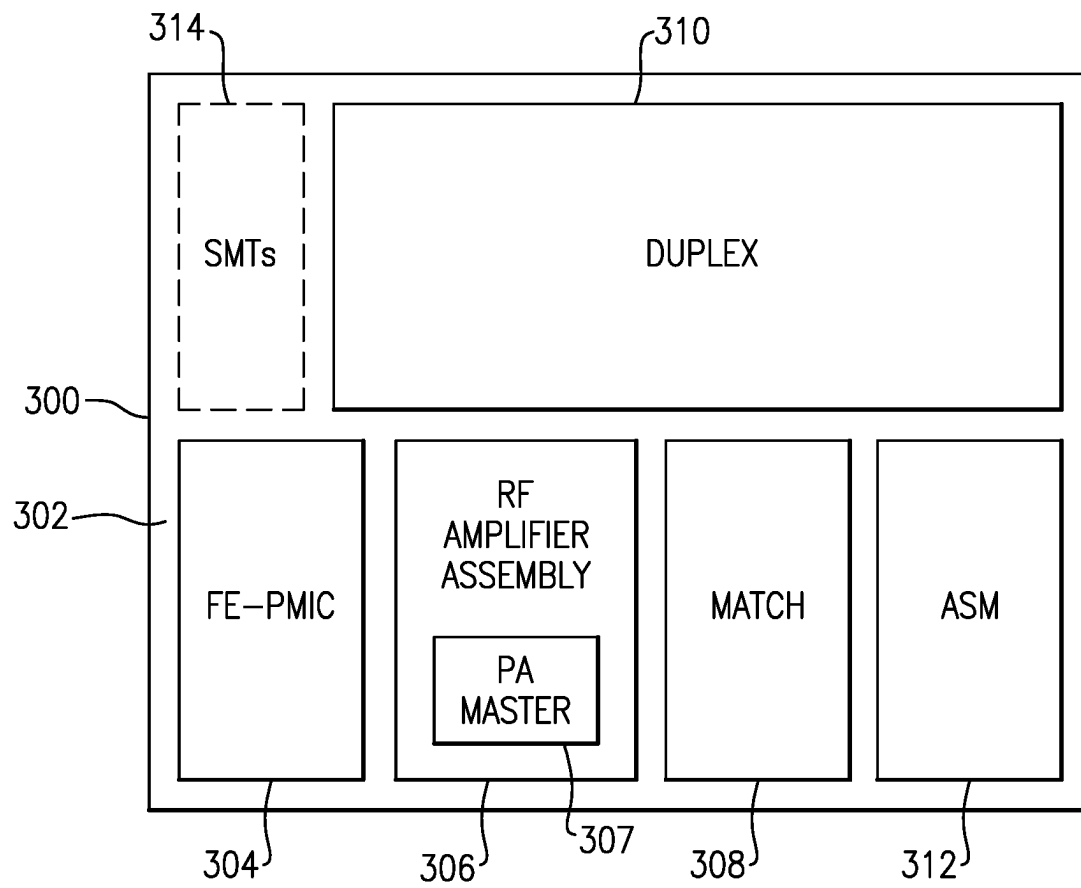
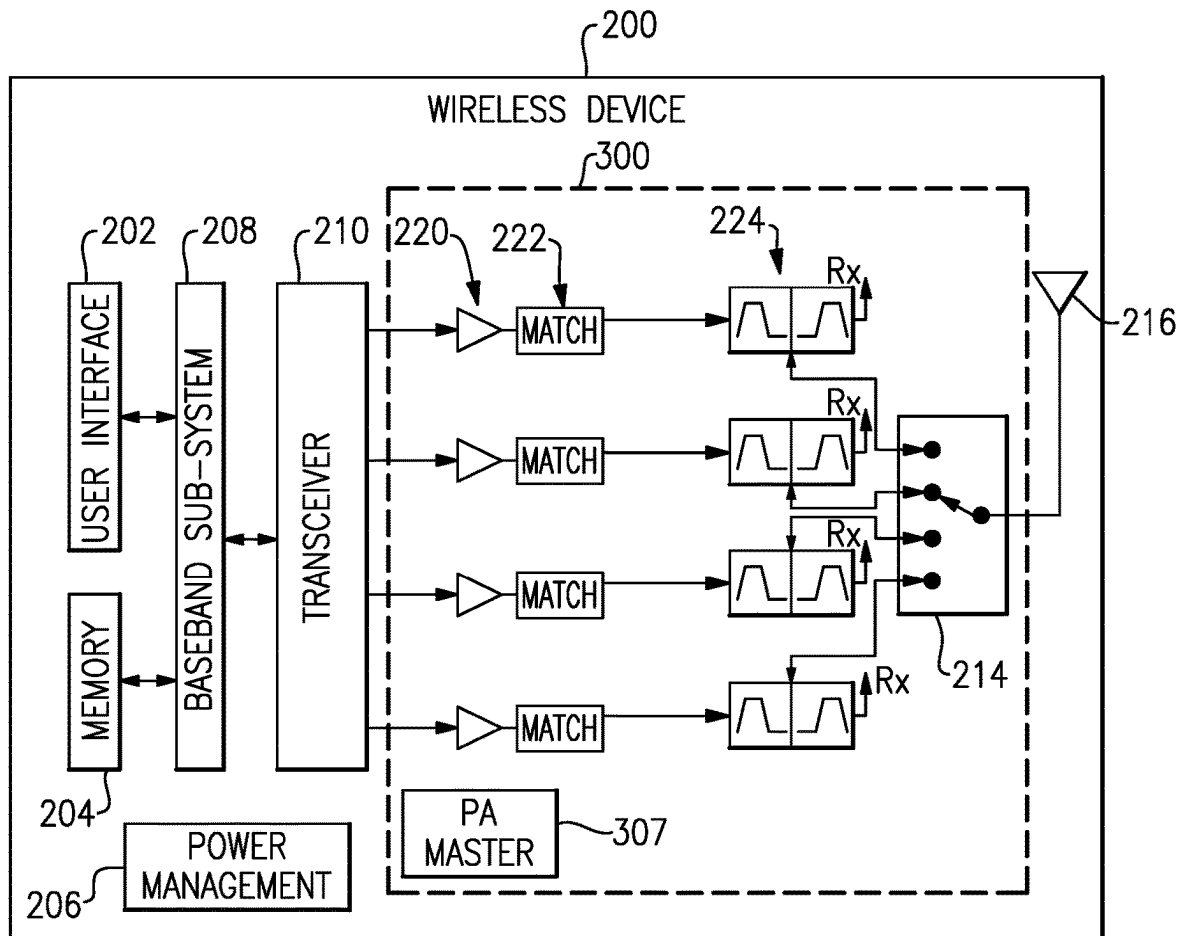


FIG.8

**FIG.9**

**FIG.10**

1

SYSTEMS AND METHODS RELATED TO POWER AMPLIFICATION AND POWER SUPPLY CONTROL

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application is a continuation of U.S. patent application Ser. No. 17/335,948, filed Jun. 1, 2021, entitled "SYSTEMS AND METHODS RELATED TO POWER AMPLIFICATION AND POWER SUPPLY CONTROL," which is a continuation of U.S. patent application Ser. No. 16/686,177, filed Nov. 17, 2019, entitled "SYSTEMS AND METHODS RELATED TO POWER AMPLIFICATION AND POWER SUPPLY CONTROL," now U.S. Pat. No. 11,025,210, issued Jun. 1, 2021, which is a continuation of U.S. patent application Ser. No. 14/867,209, filed Sep. 28, 2015, entitled "POWER AMPLIFIER MODULE WITH POWER SUPPLY CONTROL," now U.S. Pat. No. 10,483,926, issued Nov. 19, 2019, which claims priority to U.S. Provisional Application No. 62/116,463, filed Feb. 15, 2015, entitled "SWITCHING MODE POWER SUPPLY CONTROLLED BY POWER AMPLIFIER MASTER," the disclosure of each of which is hereby expressly incorporated by reference herein in its entirety.

BACKGROUND

Field

The present disclosure generally relates to power amplification systems.

Description of the Related Art

Some wireless communication devices include both a power amplifier (PA) and a switching mode power supply (SMPS). The SMPS and PA can be controlled independently by a modem/transceiver system. As a result, there is limited ability for the SMPS to be adjusted as a function of PA operating conditions.

SUMMARY

In accordance with some implementations, the present disclosure relates to a power amplification control system. The power amplification control system includes an interface configured to receive a transceiver control signal from a transceiver. The power amplification control system includes a power amplifier control component configured to generate a power amplifier control signal based on the transceiver control signal from the transceiver and a power supply control component configured to generate a power supply control signal based on the transceiver control signal from the transceiver and to generate the power supply control signal based on a local control signal from the power amplifier control component.

In some embodiments, the interface can include one or more control registers. In some embodiments, the one or more control registers can include one or more power amplifier control registers and one or more power supply control registers. In some embodiments, the power amplifier control component can be configured to generate the power amplifier control signal based on a portion of the transceiver control signal written to the one or more power amplifier control registers and the power supply control component is configured to generate the power supply control signal based

2

on a portion of the transceiver control signal written to the one or more power supply control registers. In some embodiments, the power amplifier control component can be configured to overwrite one or more of the power supply control registers with the local control signal.

In some embodiments, the power supply control component can have a first input coupled to the interface to receive at least a portion of the transceiver control signal and a second input coupled to the power amplifier control component to receive the local control signal from the power amplifier control component.

In some embodiments, the power amplifier control signal can include at least one of a bias voltage or an enable signal. In some embodiments, the power supply control signal can include at least one of a reference voltage or an enable signal.

In some embodiments, the power supply control component can be further configured to generate the power supply control signal based on an external control signal from an alternate power amplifier control component.

In some embodiments, the power amplifier control component can be configured to generate the local control signal based on a sensed condition of a power amplifier. In some embodiments, the sensed condition of the power amplifier can be at least one of a saturation condition or a safety condition.

In some implementations, the present disclosure relates to a radio-frequency (RF) module including a packaging substrate configured to receive a plurality of components. The RF module includes a power amplification system implemented on the packaging substrate. The power amplification system includes a power amplifier configured to be powered by a power supply and a control system. The control system is configured to generate a power amplifier control signal to control the power amplifier. The control system is further configured to generate a power supply control signal to control the power supply.

In some embodiments, the RF module can be a front-end module (FEM).

In some embodiments, the power supply can be implemented on the packaging substrate. In some embodiments, the power supply can include a switching mode power supply. In some embodiments, the power supply can include a boost converter.

In some embodiments, the control system can include an interface configured to receive a transceiver control signal from a transceiver, a power amplifier control component generate the power amplifier control signal based on the transceiver control signal, and a power supply control component configured to generate the power supply control signal based on the transceiver control signal from the transceiver and to generate the power supply control signal based on a local control signal from the power amplifier control component.

In some embodiments, the power supply control component can have a first input coupled to the interface to receive at least a portion of the transceiver control signal and a second input coupled to the power amplifier control component to receive the local control signal from the power amplifier control component.

In some implementations, the present disclosure relates to a wireless device including a transceiver configured to generate a radio-frequency (RF) signal. The wireless device includes a front-end module (FEM) in communication with the transceiver. The FEM includes a packaging substrate configured to receive a plurality of components. The FEM includes a power amplification system implemented on the

3

packaging substrate. The power amplification system includes a power amplifier configured to be powered by a power supply and amplify the RF signal. The power amplification system further includes a control system configured to generate a power amplifier control signal to control the power amplifier. The control system is further configured to generate a power supply control system to control the power supply. The wireless device includes an antenna in communication with the FEM. The antenna is configured to transmit the amplified RF signal.

In some embodiments, the transceiver is not directly coupled to the power supply.

For purposes of summarizing the disclosure, certain aspects, advantages and novel features of the inventions have been described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment of the invention. Thus, the invention may be embodied or carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows an example wireless system or architecture.

FIG. 2 shows that, in some implementations, an amplification system can include a radio-frequency (RF) amplifier assembly having one or more power amplifiers.

FIGS. 3A-3E show non-limiting examples of power amplifiers.

FIG. 4 illustrates a block diagram of a power amplification configuration with independent power amplifier control and power supply control.

FIG. 5 illustrates a block diagram of a power amplification configuration with integrated power amplifier control and power supply control.

FIG. 6 illustrates a block diagram of a power amplification configuration with an example controller with integrated power amplifier control and power supply control components.

FIG. 7 illustrates a block diagram of a power amplification configuration with an example controller including control registers.

FIG. 8 shows an example of how the power supply of FIG. 7 can respond to the various inputs of the reference voltage.

FIG. 9 depicts a module having one or more features as described herein.

FIG. 10 depicts a wireless device having one or more features described herein.

DETAILED DESCRIPTION OF SOME EMBODIMENTS

The headings provided herein, if any, are for convenience only and do not necessarily affect the scope or meaning of the claimed invention.

Referring to FIG. 1, one or more features of the present disclosure generally relate to a wireless system or architecture 50 having an amplification system 52. In some embodiments, the amplification system 52 can be implemented as one or more devices, and such device(s) can be utilized in the wireless system/architecture 50. In some embodiments, the wireless system/architecture 50 can be implemented in, for example, a portable wireless device. Examples of such a wireless device are described herein.

4

FIG. 2 shows that the amplification system 52 of FIG. 1 typically includes a radio-frequency (RF) amplifier assembly 54 having one or more power amplifiers (PAs). In the example of FIG. 2, three PAs 60a-60c are depicted as forming the RF amplifier assembly 54. It will be understood that other numbers of PA(s) can also be implemented. It will also be understood that one or more features of the present disclosure can also be implemented in RF amplifier assemblies having other types of RF amplifiers.

In some embodiments, the RF amplifier assembly 54 can be implemented on one or more semiconductor die, and such die can be included in a packaged module such as a power amplifier module (PAM) or a front-end module (FEM). Such a packaged module is typically mounted on a circuit board associated with, for example, a portable wireless device.

The PAs (e.g., 60a-60c) in the amplification system 52 are typically biased by a bias system 56. Further, supply voltages for the PAs are typically provided by a supply system 58. In some embodiments, either or both of the bias system 56 and the supply system 58 can be included in the foregoing packaged module having the RF amplifier assembly 54.

In some embodiments, the amplification system 52 can include a matching network 62. Such a matching network can be configured to provide input matching and/or output matching functionalities for the RF amplifier assembly 54.

For the purpose of description, it will be understood that each PA 60a-60c of FIG. 2 can be implemented in a number of ways. FIGS. 3A-3E show non-limiting examples of how such a PA can be configured. FIG. 3A shows an example PA having an amplifying transistor 64, where an input RF signal (RF_in) is provided to a base of the transistor 64, and an amplified RF signal (RF_out) is output through a collector of the transistor 64.

FIG. 3B shows an example PA having a plurality of amplifying transistors (e.g., 64a, 64b) arranged in stages. An input RF signal (RF_in) is provided to a base of the first transistor 64a, and an amplified RF signal from the first transistor 64a is output through its collector. The amplified RF signal from the first transistor 64a is provided to a base of the second transistor 64b, and an amplified RF signal from the second transistor 64b is output through its collector to thereby yield an output RF signal (RF_out) of the PA.

In some embodiments, the foregoing example PA configuration of FIG. 3B can be depicted as two or more stages as shown in FIG. 3C. The first stage 64a can be configured as, for example, a driver stage; and the second stage 64b can be configured as, for example, an output stage.

FIG. 3D shows that in some embodiments, a PA can be configured as a Doherty PA. Such a Doherty PA can include amplifying transistors 64a, 64b configured to provide carrier amplification and peaking amplification of an input RF signal (RF_in) to yield an amplified output RF signal (RF_out). The input RF signal can be split into the carrier portion and the peaking portion by a splitter. The amplified carrier and peaking signals can be combined to yield the output RF signal by a combiner.

FIG. 3E shows that in some embodiments, a PA can be implemented in a cascode configuration. An input RF signal (RF_in) can be provided to a base of the first amplifying transistor 64a operated as a common emitter device. The output of the first amplifying transistor 64a can be provided through its collector and be provided to an emitter of the second amplifying transistor 64b operated as a common base device. The output of the second amplifying transistor 64b can be provided through its collector so as to yield an amplified output RF signal (RF_out) of the PA.

5

In the various examples of FIGS. 3A-3E, the amplifying transistors are described as bipolar junction transistors (BJTs) such as heterojunction bipolar transistors (HBTs). It will be understood that one or more features of the present disclosure can also be implemented in or with other types of transistors such as field-effect transistors (FETs).

In some embodiments, the amplification system 52 of FIG. 2 can be implemented as a high-voltage (HV) power amplification system. Such a system can include an HV power amplifier assembly configured to include HV amplification operation of some or all of the PAs (e.g., 60a-60c). As described herein, such PAS can be biased by a bias system. In some embodiments, the foregoing HV amplification operation can be facilitated by an HV supply system. Such an HV supply system can include, for example a switching mode power supply (SMPS). In some embodiments, a PA master system can be implemented. Such a PA master system can include a control component for controlling some or all of the PAs. The control component can also be configured to provide at least some control of the SMPS. Although various examples are described herein in the context of HV operations, it will be understood that one or more features of the present disclosure can also be implemented in non-HV applications.

FIG. 4 illustrates a block diagram of a power amplification configuration 400 with independent power amplifier control and power supply control. The power amplification configuration 400 can be implemented in, for example, a hand-held mobile device. The power amplification configuration 400 includes a power amplifier (PA) 422 and a power supply (PS) 412. In some embodiments, the power supply 412 can include a switching mode power supply (SMPS), such as a boost converter, a buck converter, a buck-boost converter, a charge pump, etc.

The power supply 412 receives an input voltage (e.g., V_{batt} from a battery or from another source) and supplies a supply voltage (V_{cc}) to the power amplifier 422. The power amplifier 422 is powered by the supply voltage. The magnitude of the supply voltage can be set by a power supply control signal received by the power supply 412 and provided by a power supply controller 411. The power supply controller 411 can generate the power supply control signal based on a transceiver control signal received from a transceiver 401.

The power amplifier 422 receives an input signal (RF_{in}) and supplies, as an output signal (RF_{out}), an amplified version of the input signal. The input signal can be received from a power amplifier controller 421 that receives the signal from the transceiver 401 (as shown in FIG. 4), can be received directly from the transceiver 401, or can be received from another source. The power amplifier 422 is biased by power amplifier control signal (e.g., a bias signal such as a bias voltage or a bias current) received by the power amplifier 422 and provided by the power amplifier controller 421. The power amplifier controller 421 can generate the power amplifier control signal based on a transceiver control signal received from the transceiver 401. The power amplifier controller 421 and power amplifier 422 can be integrated onto a single module 420. In particular, the power amplifier controller 421 and power amplifier 422 can be integrated on a single die. In some implementations, the power supply 412 can also be integrated onto the module or integrated on the die.

Thus, the power supply 412 and the power amplifier 422 are controlled independently by the transceiver 401. As a result, there is limited ability for the power supply to be adjusted as a function of the operating conditions of the

6

power amplifier 422. In some configurations, control of the power amplifier 422 and the power supply 412 can be accomplished through fixed conditions. As an example, a system request for a specified output power drives a setup condition for the power amplifier 422 and the power supply 412 based upon a pre-determined table.

In some configurations, additional interface signals between the power amplifier 422 and the transceiver 401 (e.g., via the power amplifier controller 421) allow sensing of power amplifier outputs such that the transceiver 401 can change the operating state of the power supply 401 based upon these sensed conditions. For example, in response to a sensed condition of the power amplifier 422, the power amplifier controller 421 can send a request to the transceiver 401 to increase the supply voltage. In response, the transceiver 401 can send a transceiver control signal to the power supply controller 411 which can, in turn, send a power supply control signal to the power supply 412 to increase the supply voltage (V_{cc}).

In some embodiments, as described in detail below, some or all of the power supply control features can be integrated with some or all of the power amplifier control features. Such an integration can allow the power supply to respond to inputs from the transceiver, from the power amplification module directly, or any combination thereof. For example, internal monitoring within the power amplifier control function can generate one or more outputs which can be utilized to directly control the power supply and adjust the supply voltage applied to the power amplifier without any required knowledge by the transceiver.

Integration of power supply and power amplifier control features can provide significant advantages. For example, customizations for the transceiver system or software are not required in order to support more advanced features associated with supplying of power to power amplifier. In some embodiments, traditional setup conditions can be provided by the transceiver, while the power supply can then be adjusted based upon one or more specific parameters required or desired by the associated power amplifier.

FIG. 5 illustrates a block diagram of a power amplification configuration 500 with integrated power amplifier control and power supply control. Such integration can allow, for example, interaction between the power amplifier 522 and power supply 512 without input from or interaction with the transceiver 501. In particular, the transceiver 501 is not directly coupled to the power supply 512 and is not coupled to the power supply 512 via a dedicated power supply controller.

The power amplification configuration 500, like the configuration 400 of FIG. 4, includes a power amplifier 522 and a power supply 512. In some embodiments, the power supply 512 can include a switching mode power supply (SMPS), such as a boost converter, a buck converter, a buck-boost converter, a charge pump, etc.

The power supply 512 receives an input voltage (e.g., V_{batt} from a battery or from another source) and supplies a supply voltage (V_{cc}) to the power amplifier 522. The power amplifier 522 is powered by the supply voltage. The magnitude of the supply voltage can be set by a power supply control signal received by the power supply 512 and provided by an integrated controller 521. The controller 521 can generate the power supply control signal based on a transceiver control signal received from a transceiver 501 or based on a sensed condition of the power amplifier 522.

The power amplifier 522 receives an input signal (RF_{in}) and supplies, as an output signal (RF_{out}), an amplified version of the input signal. The input signal can be received

from the controller 521 that receives the signal from the transceiver 501 (as shown in FIG. 5), can be received directly from the transceiver 501, or can be received from another source. The power amplifier 522 is biased by power amplifier control signal (e.g., a bias signal such as a bias voltage or a bias current) received by the power amplifier 522 and provided by the controller 521. The power amplifier controller 521 can generate the power amplifier control signal based on a transceiver control signal received from the transceiver 501. The controller 521 and power amplifier 522 can be integrated onto a single module 520, referred to herein as a PA master. In particular, the controller 521 and power amplifier 522 can be integrated on a single die. In some implementations, the power supply 512 can also be integrated onto the module or integrated on the die.

FIG. 6 illustrates a block diagram of a power amplification configuration 600 with an example controller 621 with integrated power amplifier control and power supply control components 632, 633. The power amplification configuration 600, like the configuration 500 of FIG. 5, includes a power amplifier 622 and a power supply 612. In some embodiments, the power supply 612 can include a switching mode power supply (SMPS), such as a boost converter, a buck converter, a buck-boost converter, a charge pump, etc.

The power supply 612 receives an input voltage (e.g., Vbatt from a battery or from another source) and supplies a supply voltage (Vcc) to the power amplifier 622. The power amplifier 622 is powered by the supply voltage. The magnitude of the supply voltage can be set by a power supply control signal received by the power supply 612 and provided by a controller 621. In particular, the power supply control signal can be provided by the power supply control component 632 of the controller 621. The power supply control component 632 can generate the power supply control signal based on a transceiver control signal received from a transceiver 501 via an interface 631 or based on local control signal received from the power amplifier control component 633. The local control signal can be, for example, based on sensed condition of the power amplifier 622. As shown in FIG. 6, the power supply control component 632 includes a first input coupled to the interface 631 to receive at least a portion of the transceiver control signal and a second input coupled to the power amplifier control component 633 to receive the local control signal from the power amplifier control component 633.

The power amplifier 622 receives an input signal (RF_{in}) and supplies, as an output signal (RF_{out}), an amplified version of the input signal. The input signal can be received from the power amplifier control component 633 that receives the signal from the transceiver 601 via the interface 631 (as shown in FIG. 6), can be received directly from the transceiver 601, or can be received from another source. The power amplifier 622 is biased by power amplifier control signal (e.g., a bias signal such as a bias voltage or a bias current) received by the power amplifier 622 and provided by the power amplifier control component 633. The power amplifier control component 633 can generate the power amplifier control signal based on a transceiver control signal received from the transceiver 601 via the interface 631. The controller 621 and power amplifier 622 can be integrated onto a single module 620, referred to herein as a PA master. In particular, the controller 621 and power amplifier 622 can be integrated on a single die. In some implementations, the power supply 612 can also be integrated onto the module or integrated on the die.

Thus, the power amplification configuration 600 of FIG. 6 includes a power amplification control system including

the controller 621. The controller 621 includes an interface 631 configured to receive a transceiver control signal from the transceiver 601. The controller 621 includes a power amplifier control component 633 configured to generate a power amplifier control signal based on the transceiver control signal from the transceiver 601 and a power supply control component 632 configured to generate a power supply control signal based on the transceiver control signal from the transceiver 601 and to generate the power supply control signal based on a local control signal from the power amplifier control component 633.

In some embodiments, the power amplifier control signal supplied to the power amplifier 622 by the power amplifier control component 633 includes a bias voltage for biasing the power amplifier 622. In some embodiments, the power amplifier control signal includes an enable signal for enabling (or disabling) the power amplifier 622.

In some embodiments, the power supply control signal supplied to the power supply 612 by the power supply control component 632 includes a reference voltage that indicates a magnitude of the supply voltage to be provided to the power amplifier 622. In some embodiments, the power supply control signal includes an enable signal for enabling (or disabling) the power supply 612.

In some embodiments, the local control signal supplied to the power supply control component 632 by the power amplifier control component 633 indicates that the supply voltage is to be increased. In some embodiments, the local control signal indicates that the supply voltage is to be decreased. In some embodiments, the local control signal indicates that the power supply 612 is to be disabled.

In some embodiments, the local control signal is based on a sensed condition of the power amplifier 622. The sensed condition can be a saturation condition or a safety condition. For example, the power amplifier control component 633 can detect that the power amplifier 622 (or one or more transistors of the power amplifier 622) is saturated. In response, the power amplifier control component 633 can provide a local control signal to the power supply control component 632 indicating that the supply voltage is to be increased. As another example, the power amplifier control component 633 can detect that the power amplifier 622 is operating in (or approaching operation in) an unsafe condition that could lead to damage to the power amplifier 622. In response, the power amplifier control component 633 can provide a local control signal to the power supply control component 632 indicating that the supply voltage is to be decreased or the power supply 612 is to be disabled.

FIG. 7 illustrates a block diagram of a power amplification configuration 700 with an example controller 721 including control registers 731. The power amplification configuration 700 includes a power amplifier 722 and a switching mode power supply (SMPS) 712. The switching mode power supply can include a boost converter, a buck converter, a buck-boost converter, a charge pump, etc. The power amplifier 722 can be a high-voltage power amplifier.

The SMPS 712 receives an input voltage (e.g., from a battery or from another source) and supplies a supply voltage at an output to the power amplifier 722. The power amplifier 722 is powered by the supply voltage. The magnitude of the supply voltage can be set by a power supply control signal received by the power supply 712 and provided by a controller 721. In particular, the power supply control signal can be provided by an SMPS control component 732 of the controller 721. The power supply control signal can be converted from a digital signal to an analog reference voltage (V_{ref}) by a digital-to-analog converter

734. The SMPS control component **732** can generate the power supply control signal based on a transceiver control signal received from a modem/transceiver (not shown) via an interface **731** or based on local control signal received from a power amplifier bias control component **733**. The local control signal can be, for example, based on sensed condition of the power amplifier **722**. As shown in FIG. 7, the SMPS control component **732** includes a first input coupled to the interface **731** to receive at least a portion of the transceiver control signal and a second input coupled to the power amplifier bias control component **733** to receive the local control signal from the power amplifier bias control component **733**.

The SMPS control component **732** can also generate the power supply control signal based on an external control signal received from an alternate PA module **750** including a power amplifier power by the supply voltage of the SMPS **712**. In particular, the external control signal can be received from a power amplifier control component (e.g., a power amplifier bias control component) of the alternate PA module **750**.

The power amplifier **722** receives an input signal (RF_{in}) and supplies, as an output signal (RF_{out}), an amplified version of the input signal. The input signal can be received from the power amplifier bias control component **733** that receives the signal from the transceiver via the interface **731** (as shown in FIG. 7), can be received directly from the transceiver, or can be received from another source (such as another power amplifier control component). The power amplifier **722** is biased by power amplifier control signal (e.g., a bias signal such as a bias voltage or a bias current) received by the power amplifier **722** and provided by the power amplifier bias control component **733**. The power amplifier bias control component **733** can generate the power amplifier control signal based on a transceiver control signal received from the transceiver via the interface **731**. The controller **721** and power amplifier **722** can be integrated onto a single module **720**, referred to herein as a PA master. In particular, the controller **721** and power amplifier **722** can be integrated on a single die. In some implementations, the power supply **712** can also be integrated onto the module or integrated on the die.

The interface **731** of the controller **721** includes one or more control registers. The control registers can be, for example, MIPI® control registers. In particular, as shown in FIG. 7, the interface **731** includes one or more power amplifier control registers and one or more power supply control registers. The interface **731** further includes one-time programmable (OTP) memory.

The power amplifier bias control component **733** is configured to generate the power amplifier control signal based on a portion of the transceiver control signal written to the one or more power amplifier control registers and the SMPS control component **732** is configured to generate the power supply control signal based on a portion of the transceiver control signal written to the one or more power supply control registers. In some embodiments, the power amplifier bias control component **733** is configured to overwrite one or more of the power supply control registers with the local control signal. Thus, the local control signal can be provided, in some implementations, to the SMPS control component **732** via the interface **731**.

The interface **731** can include an input/output voltage (VIO) pin, a clock (CLK) pin, a ground (GND) pin, and a data pin. The transceiver control signal can be transmitted from the modem/transceiver (and written to the control registers) via the data pin.

Thus, the power amplification configuration **700** of FIG. 7 includes a power amplification control system including the controller **721**. The controller **721** includes an interface **731** configured to receive a transceiver control signal from the transceiver. The controller **721** includes a power amplifier control component (e.g., the power amplifier bias control component **733**) configured to generate a power amplifier control signal (e.g., a bias voltage) based on the transceiver control signal from the transceiver and a power supply control component (e.g., the SMPS control component **732**) configured to generate a power supply control signal (e.g., the reference voltage) based on the transceiver control signal from the transceiver and to generate the power supply control signal based on a local control signal from the power amplifier control component.

Table 1 lists examples of control signals that can be generated by the SMPS control component **732** by using a 3-bit signal (which can be written to one of the SMPS control registers) to generate various values of V_{ref}. Table 1 illustrates that a number of operating modes can be implemented with varying values of V_{ref}, including a disable mode. In the “Boost” mode, the specific V_{ref} output can be indicated using a value written to another of the SMPS control registers.

TABLE 1

Control B2	Control B1	Control Bit B0	Mode	Vref output
0	0	0	Disable	0
0	1	0	Forced Bypass	0.6
0	1	1	Boost	>0.8
1	0	0	2 G Bias	<0.3
1	0	1	Reserved	NA
1	1	0	Reserved	NA
1	1	1	Reserved	NA

FIG. 8 shows an example of how the SMPS **712** of FIG. 7 can respond to the various inputs of the reference voltage V_{ref}, some of which are listed in Table 1. In the 2G bias mode (V_{ref}<0.3V), the SMPS can output a voltage suitable for 2G biasing purpose. In the buck mode (V_{ref} between 0.3V and 0.4V), the SMPS can output a voltage that is, for example, half of the battery voltage. In the bypass mode (V_{ref} between 0.4V and 0.8V), the SMPS can output a voltage that is substantially equal to the battery voltage. In the boost mode (V_{ref}>0.8V), the SMPS can output a boosted voltage that is proportional to V_{ref}. Such an output can be utilized as, for example, a supply voltage for one or more PAs operating in HV mode.

FIG. 9 shows that in some embodiments, some or all of the integrated PA master functionalities described herein can be implemented in a module. Such a module can be, for example, a front-end module (FEM). In the example of FIG. 9, a module **300** can include a packaging substrate **302**, and a number of components can be mounted on such a packaging substrate. For example, an FE-PMIC component **302**, a power amplifier assembly **306** including a PA master **307**, a match component **308**, and a duplexer assembly **310** can be mounted and/or implemented on and/or within the packaging substrate **302**. Other components such as a number of SMT devices **314** and an antenna switch module (ASM) **312** can also be mounted on the packaging substrate **302**. Although all of the various components are depicted as being laid out on the packaging substrate **302**, it will be understood that some component(s) can be implemented over other component(s).

11

In some implementations, a device and/or a circuit having one or more features described herein can be included in an RF device such as a wireless device. Such a device and/or a circuit can be implemented directly in the wireless device, in a modular form as described herein, or in some combination thereof. In some embodiments, such a wireless device can include, for example, a cellular phone, a smart-phone, a hand-held wireless device with or without phone functionality, a wireless tablet, etc.

FIG. 10 depicts an example wireless device **200** having one or more advantageous features described herein. In the context of a module having one or more features as described herein, such a module can be generally depicted by a dashed box **300**, and can be implemented as, for example, a front-end module (FEM). Such a module can include a PA master **307** having one or more features as described herein.

Referring to FIG. 10, power amplifiers (PAs) **220** can receive their respective RF signals from a transceiver **210** that can be configured and operated in known manners to generate RF signals to be amplified and transmitted, and to process received signals. The transceiver **210** is shown to interact with a baseband sub-system **208** that is configured to provide conversion between data and/or voice signals suitable for a user and RF signals suitable for the transceiver **210**. The transceiver **210** can also be in communication with a power management component **206** that is configured to manage power for the operation of the wireless device **200**. Such power management can also control operations of the baseband sub-system **208** and the module **300**.

The baseband sub-system **208** is shown to be connected to a user interface **202** to facilitate various input and output of voice and/or data provided to and received from the user. The baseband sub-system **208** can also be connected to a memory **204** that is configured to store data and/or instructions to facilitate the operation of the wireless device, and/or to provide storage of information for the user.

In the example wireless device **200**, outputs of the PAs **220** are shown to be matched (via respective match circuits **222**) and routed to their respective duplexers **220**. Such amplified and filtered signals can be routed to an antenna **216** through an antenna switch **214** for transmission. In some embodiments, the duplexers **220** can allow transmit and receive operations to be performed simultaneously using a common antenna (e.g., **216**). In FIG. 10, received signals are shown to be routed to “Rx” paths (not shown) that can include, for example, a low-noise amplifier (LNA).

A number of other wireless device configurations can utilize one or more features described herein. For example, a wireless device does not need to be a multi-band device. In another example, a wireless device can include additional antennas such as diversity antenna, and additional connectivity features such as Wi-Fi, Bluetooth, and GPS.

Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” The word “coupled”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Description using the singular or plural number may also include the plural or singular number

12

respectively. The word “or” in reference to a list of two or more items, that word covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

The above detailed description of embodiments of the invention is not intended to be exhaustive or to limit the invention to the precise form disclosed above. While specific embodiments of, and examples for, the invention are described above for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative embodiments may perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified. Each of these processes or blocks may be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks may instead be performed in parallel, or may be performed at different times.

The teachings of the invention provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various embodiments described above can be combined to provide further embodiments.

While some embodiments of the inventions have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

What is claimed is:

1. A power amplification control system comprising:

an interface configured to receive a transceiver control signal from a transceiver, the interface including one or more control registers including one or more power amplifier control registers and one or more power supply control registers; and

a control system configured to generate a power amplifier control signal based on the transceiver control signal from the transceiver and configured to generate a local control signal, the control system configured to overwrite one or more of the power supply control registers with the local control signal, the control system further configured to generate a power supply control signal based on one or more of the transceiver control signal and the local control signal.

2. The power amplification control system of claim 1 wherein the control system is configured to generate the power amplifier control signal based on a portion of the transceiver control signal written to the one or more power amplifier control registers and the control system is configured to generate the power supply control signal based on a portion of the transceiver control signal written to the one or more power supply control registers.

3. The power amplification control system of claim 1 wherein the power amplifier control signal includes at least one of a bias voltage or an enable signal.

4. The power amplification control system of claim 1 wherein the power supply control signal includes at least one of a reference voltage or an enable signal.

13

5. The power amplification control system of claim 1 wherein the local control signal is based on a sensed condition of a power amplifier.

6. The power amplification control system of claim 5 wherein the sensed condition of the power amplifier is at least one of a saturation condition or a safety condition.

7. The power amplification control system of claim 1 wherein the control system includes a power amplifier control component and a power supply control component, the power amplifier control component configured to generate the power amplifier control signal and the local control signal, and the power supply control component configured to generate the power supply control signal.

8. The power amplification control system of claim 7 wherein the power supply control component has a first input coupled to the interface to receive at least a portion of the transceiver control signal and a second input coupled to the power amplifier control component to receive the local control signal from the power amplifier control component.

9. The power amplification control system of claim 1 wherein the control system is further configured to generate the power supply control signal based on an external control signal from an alternate power amplifier control component.

10. A method of implementing a radio-frequency module, the method comprising:

providing a packaging substrate configured to receive a plurality of components; and

providing a power amplification system implemented on the packaging substrate, the power amplification system including a power amplifier configured to be powered by a power supply and a control system, the control system including an interface configured to receive a transceiver control signal from a transceiver, the interface including one or more control registers including one or more power amplifier control registers and one or more power supply control registers, the control system further configured to generate a power amplifier control signal to control the power amplifier based on the transceiver control signal from the transceiver and configured to generate a local control signal, the control system configured to overwrite one or more of the power supply control registers with the local control signal, the control system further configured to

14

generate a power supply control signal to control the power supply based on one or more of the transceiver control signal and the local control signal.

11. The method of claim 10 wherein the radio-frequency module is a front-end module.

12. The method of claim 10 wherein the power supply is implemented on the packaging substrate.

13. The method of claim 10 wherein the power supply includes a switching mode power supply.

14. The method of claim 10 wherein the power supply includes a boost converter.

15. The method of claim 10 wherein the control system further includes a power amplifier control component and a power supply control component, the power amplifier control component configured to generate the power amplifier control signal and the local control signal, and the power supply control component configured to generate the power supply control signal.

16. The method of claim 15 wherein the power supply control component has a first input coupled to the interface to receive at least a portion of the transceiver control signal and a second input coupled to the power amplifier control component to receive the local control signal from the power amplifier control component.

17. The method of claim 15 wherein the power amplifier control component is configured to generate the power amplifier control signal based on a portion of the transceiver control signal written to the one or more power amplifier control registers and the power supply control component is configured to generate the power supply control signal based on a portion of the transceiver control signal written to the one or more power supply control registers.

18. The method of claim 10 wherein the power amplifier control signal includes at least one of a bias voltage or an enable signal.

19. The method of claim 10 wherein the power supply control signal includes at least one of a reference voltage or an enable signal.

20. The method of claim 10 wherein the control system is further configured to generate the power supply control signal based on an external control signal from an alternate power amplifier control component.

* * * * *